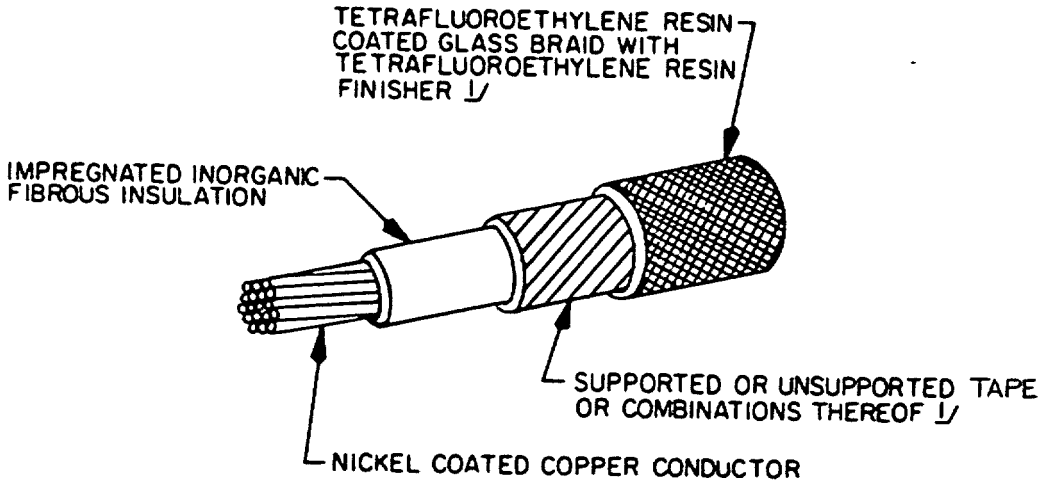
MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, HIGH TEMPERATURE AND FIRE RESISTANT, NORMAL WEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the latest issue of Specification MIL-W-25038.



1/ The outer two layers may be reversed in their sequence. The reversal is optional.

FIGURE 1. Wire, electrical.

REQUIREMENTS:

Design and construction: See figure 1.

Temperature rating: 260°C.

Voltage rating: 600 Vrms.

Color: Unless otherwise specified by the part number, the color shall be white.

Thermal shock: Maximum change in insulation measurement:
0.060 inch for sizes 22 through 10;
0.125 inch for sizes 8 through 1;
0.187 inch for sizes 0 through 0000.

Durability of identification: 250 cycles (500 strokes).

Part number: M25038/1- (wire size from table I) - (color code number).
Examples:
Size 20, white - M25038/1-20-9;
Size 20, white with orange stripe - M25038/1-20-93.

FSC 6145

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TABLE I. Finished wire construction and performance details.

Finished wire construction				Performance details	
				Bend test and cold bend	
Wire size	Resistance at 20°C (68°F) ohms/1000 ft (max)	Diameter in inches	Weight lb/1000 ft (max)	Mandrel diameter in inches	Test load (lb)
22	23.7	.108 ±.008	10	6	2
20	14.6	.117 ±.008	12	6	2
18	9.14	.127 ±.008	15	6	2
16	6.85	.137 ±.010	19	8	3
14	4.32	.160	25	8	3
12	2.78	.175	35	8	3
10	1.68	.220	55	10	5
8	.936	.268 ±.012	85	12	5
6	.591	.330	127	14	10
4	.375	.395	192	16	10
2	.241	.472	291	20	15
1	.196	.515 ±.018	347	22	15
0	.153	.555 ±.018	415	24	20
00	.120	.615 ±.020	520	24	25
000	.096	.680 ±.020	648	24	30
0000	.077	.750 ±.020	793	24	30

Custodians:
Army - CR
Navy - AS
Air Force - 11

Review activities:
Army - MI, AR
Air Force - 85, 99
DLA - IS

User activities:
Army - AV
Navy - OS

Preparing activity:
Air Force - 11

Agent:
DLA - ES

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